ASSIGNMENT 11

Find Exposed Android App Components and Assess Security Weaknesses

1. Introduction

Android applications contain components such as Activities, Services, Broadcast Receivers, and Content Providers. These components allow the application to display screens, perform background tasks, respond to events, and share data. If a component is unnecessarily exported or lacks suitable access controls, another application may be able to invoke it or access data in unintended ways. This practical explains how to inspect an authorized Android application for exposed components and assess potential security weaknesses.

2. Objectives

- Understand Android application components and the meaning of component exposure.
- Inspect an authorized APK or test application for exported components.
- Review manifest settings and component permissions.
- Identify potential risks and recommend secure configurations.

3. Requirements

- An Android test device or emulator
- An application APK that you own or are authorized to assess
- Android SDK tools, including ADB and optionally APK Analyzer
- A computer with Android Studio or equivalent inspection tools

Safety note: Assess only your own application or an APK for which you have explicit permission. Do not access other users' data or attempt to bypass protections on third-party applications.

4. Understanding Exposed Components

An Android component is considered exported when it can be accessed by other applications, subject to Android's rules and any required permissions. Exported components are not always vulnerable; some must be public to support legitimate app functionality. The key question is whether exposure is necessary and adequately protected.

Component	Purpose	Potential Concern
Activity	Provides a user-facing screen.	A sensitive screen may be launched without expected authorization.
Service	Performs background work.	An external app may invoke an unprotected operation.
Broadcast Receiver	Responds to broadcast	An untrusted sender may

	messages.	trigger unintended actions.
Content Provider	Shares structured data.	Insufficient access control may expose private data.

5. Inspecting the Android Manifest

The AndroidManifest.xml file declares application components and their configuration. In Android Studio, open the APK with APK Analyzer or inspect the source project's manifest. Review each Activity, Service, Receiver, and Provider, paying attention to android:exported, intent filters, and permission requirements.

```
<activity android:name=".AdminActivity" android:exported="false" />
```

This example shows a sensitive Activity configured as non-exported. For components that genuinely need external access, use the minimum necessary permissions and validate all incoming data.

6. Using ADB to Review an Installed Test App

Connect an authorized Android device or emulator with USB debugging enabled, or use the emulator provided by Android Studio. Confirm that ADB can see the test device:

```
adb devices
```

```
adb shell pm list packages -3
```

The first command lists connected devices, and the second lists third-party package names. To review package information for an app you are authorized to test, run the following command after replacing the placeholder with its package identifier:

```
adb shell dumpsys package <package.name>
```

Review the output for component declarations and relevant permission information. Output varies by Android version; use the manifest or APK Analyzer as the primary source for declared component settings. Do not attempt to invoke sensitive third-party components.

7. Assessing Security Weaknesses

- Unnecessary export: A component is available to other applications even though it does not need external access.
- Missing permission checks: An exported component performs sensitive actions without suitable authorization.
- Untrusted input: A component accepts Intent extras, URIs, or other input without validation.
- Data exposure: A Content Provider exposes sensitive records or allows broader access than required.
- Insecure broadcasts: A receiver accepts untrusted broadcasts that can trigger privileged behavior.

Treat these as potential weaknesses until they are confirmed by reviewing the application's intended behavior and configuration. An exported component alone does not prove a vulnerability.

8. Security Recommendations

- Set android:exported="false" for components that do not require access from other applications.
- For components that must be exported, enforce suitable permissions and authentication or authorization checks.
- Validate all incoming Intent data, extras, and content URIs.
- Limit Content Provider access and avoid exposing sensitive data unnecessarily.
- Protect sensitive operations on the server or data layer, not only in the user interface.
- Retest the manifest and application behavior after making security changes.

9. Results and Observation Table

Record the results from the authorized APK or test application. Do not report a weakness unless the configuration or behavior supports the finding.

Component	Exported?	Permission / Protection	Assessment
Activity	Record finding	Record finding	No issue / Review needed
Service	Record finding	Record finding	No issue / Review needed
Broadcast Receiver	Record finding	Record finding	No issue / Review needed
Content Provider	Record finding	Record finding	No issue / Review needed

10. Conclusion

This practical demonstrates how to identify Android application components and review whether they are exposed to other applications. By examining the manifest, exported settings, permissions, and input handling, a tester can identify potential weaknesses involving unauthorized component access or data exposure. Restricting unnecessary exports, enforcing permissions, and validating untrusted input help reduce the application's attack surface. Testing should be limited to apps and devices for which authorization has been granted.

References

1. Android Developers- App manifest overview: https://developer.android.com/guide/topics/manifest/manifest-intro
2. Android Developers - Activity element: https://developer.android.com/guide/topics/manifest/activity-element
3. Android Developers- Security best practices: https://developer.android.com/privacy-and-security/security-best-practices